NAME ______________________ DATE ______________________ PERIOD __________

Paper Practice 2.7 • **WORKED KEY**

Using Formulas

WORKED KEY (odd-numbered items) – Treat the other letters like numbers: undo what is done to the variable you want, and leave your answer as an expression. Even-numbered items are yours to work.

Examples 1 and 2

Solve each equation or formula for the variable indicated.

1. $2a - 3b = 4$, for b

Answer: $-3b = 4 - 2a \rightarrow \boldsymbol{b = \frac{2a - 4}{3}}$

2. $-m + 5n = -2$, for n

3. $6c + 2d = 7$, for c

Answer: $6c = 7 - 2d \rightarrow \boldsymbol{c = \frac{7 - 2d}{6}}$

4. $5x - 8y = 1$, for y

5. $t = 7g$, for g

Answer: $\boldsymbol{g = \frac{t}{7}}$

6. $v + 8w = 4v$, for v

7. $9p - 3q = 4q$, for q

Answer: $9p = 7q \rightarrow \boldsymbol{q = \frac{9p}{7}}$

8. $10s + 6t = 3s$, for s

9. $p = st + r$, for t

Answer: $p - r = st \rightarrow \boldsymbol{t = \frac{p - r}{s}}$

10. $2a - b = 4c$, for a

11. $2j = \frac{2}{5}k - m$, for k

Answer: $2j + m = \frac{2}{5}k \rightarrow$ multiply by $\frac{5}{2}$: $\boldsymbol{k = \frac{5}{2}(2j + m) = 5j + \frac{5}{2}m}$

12. $6x - \frac{7}{9}y = z$, for y

13. $\frac{5cd + b}{8} = 2$, for d

Answer: $5cd + b = 16 \rightarrow 5cd = 16 - b \rightarrow \boldsymbol{d = \frac{16 - b}{5c}}$

14. $\frac{pq + 7}{2} = n$, for p

Example 3

15. TEMPERATURE The formula $C = \frac{Wtc}{1000}$ represents the cost C in cents to operate an electrical device, where W is the wattage of the device, t is the time in hours that the device is in use, and c is the cost in cents per kilowatt-hour.

a. Solve the formula for W.

b. If the cost to operate a device for 5 hours is \$0.1875 and the cost per kilowatt hour is \$0.15, find the wattage of the device.

Answer: a. $1000C = Wtc \rightarrow \boldsymbol{W = \frac{1000C}{tc}}$ **b.** the formula is in cents: $C = 18.75$¢, $c = 15$¢/kWh, $t = 5 \rightarrow W = \frac{1000(18.75)}{5(15)} = \frac{18750}{75} =$ **250 watts**

16. VOLUME The formula for the volume of a cone is $V = \frac{1}{3}\pi r^2 h$, where V is the volume, r is the radius of the base, and h is the height.

a. Solve the formula for h.

b. What is the height of a cone with a volume of 1017.36 square inches and a radius of 6 inches? Use 3.14 to approximate π.

Example 4

17. **SHOPPING** Sang buys hats and T-shirts as gifts for his friends while on vacation. The total cost C in dollars of buying h hats and t T-shirts is $C = 10h + 15t$.

a. Sang has \$120 to spend on gifts. Describe the constraints.

b. Solve for h.

c. He needs to buy 4 T-shirts. How many hats can he buy?

Answer: a. $10h + 15t \le 120$, with $h \ge 0$ and $t \ge 0$ whole numbers. **b.** $10h = C - 15t \rightarrow \boldsymbol{h = \frac{C - 15t}{10}}$ **c.** $t = 4$: $h = \frac{120 - 60}{10} =$ **6 hats** (at most 6; check: $10(6) + 15(4) = 120$ ✓)